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Subject: Programming with Java

ASSIGNMENT No : 18

TITLE : Java GUI Button Event Handling

Problem Statement

Write a Java GUI program to demonstrate button click event handling.

Learning Objectives

To implement button click events using Java event handling mechanisms.

Programme

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;

public class Calculator extends JFrame implements ActionListener {

    JLabel label1, label2, resultLabel;
    JTextField num1, num2, result;
    JButton addButton, subButton;

    Calculator() {
```

```
setTitle("Simple Calculator");
setSize(400, 250);
setLayout(new GridLayout(4, 2, 10, 10));
setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

// Labels
label1 = new JLabel("Enter First Number:");
label2 = new JLabel("Enter Second Number:");
resultLabel = new JLabel("Result:");

// Text fields
num1 = new JTextField();
num2 = new JTextField();
result = new JTextField();

result.setEditable(false);

// Buttons
addButton = new JButton("Addition");
subButton = new JButton("Subtraction");

// Add action listeners
addButton.addActionListener(this);
subButton.addActionListener(this);

// Add components
add(label1);
add(num1);

add(label2);
add(num2);

add(addButton);
add(subButton);
```

```
        add(resultLabel);
        add(result);

        setVisible(true);
    }

    public void actionPerformed(ActionEvent e) {

        try {
            double a = Double.parseDouble(num1.getText());
            double b = Double.parseDouble(num2.getText());

            if (e.getSource() == addButton) {
                result.setText(String.valueOf(a + b));
            }

            if (e.getSource() == subButton) {
                result.setText(String.valueOf(a - b));
            }

        } catch (NumberFormatException ex) {
            JOptionPane.showMessageDialog(
                this,
                "Please enter valid numbers."
            );
        }
    }

    public static void main(String[] args) {
        new Calculator();
    }
}
```

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;

public class BankBalanceCalculator extends JFrame implements
ActionListener {

    JLabel initialLabel, transactionLabel, balanceLabel;
    JTextField initialField, transactionField, balanceField;
    JButton depositButton, withdrawButton;

    double balance;

    BankBalanceCalculator() {

        setTitle("Bank Balance Calculator");
        setSize(450, 300);
        setLayout(new GridLayout(4, 2, 10, 10));
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

        // Labels
        initialLabel = new JLabel("Initial Balance:");
        transactionLabel = new JLabel("Transaction Amount:");
        balanceLabel = new JLabel("Updated Balance:");

        // Text fields
        initialField = new JTextField();
        transactionField = new JTextField();
        balanceField = new JTextField();

        balanceField.setEditable(false);

        // Buttons
```

```
depositButton = new JButton("Deposit");
withdrawButton = new JButton("Withdraw");

// Add action listeners
depositButton.addActionListener(this);
withdrawButton.addActionListener(this);

// Add components
add(initialLabel);
add(initialField);

add(transactionLabel);
add(transactionField);

add(depositButton);
add(withdrawButton);

add(balanceLabel);
add(balanceField);

setVisible(true);
}

public void actionPerformed(ActionEvent e) {

    try {
        double initialBalance =
            Double.parseDouble(initialField.getText());

        double amount =
            Double.parseDouble(transactionField.getText());

        // Deposit
        if (e.getSource() == depositButton) {
```

```

        balance = initialBalance + amount;
    }

    // Withdrawal
    if (e.getSource() == withdrawButton) {
        balance = initialBalance - amount;
    }

    // Display updated balance
    balanceField.setText(String.valueOf(balance));

} catch (NumberFormatException ex) {

    JOptionPane.showMessageDialog(
        this,
        "Please enter valid numbers.",
        "Error",
        JOptionPane.ERROR_MESSAGE
    );
}

}

public static void main(String[] args) {
    new BankBalanceCalculator();
}
}

```

Output

Simple Calculator

Enter First Number:

20

Enter Second Number:

10

Addition

Subtraction

Result:

30.0

Bank Balance Calculator

Initial Balance:

12000

Transaction Amount:

2000

Deposit

Withdraw

Updated Balance:

14000.0

Conclusion

The Java GUI programs successfully demonstrated **button click event handling using ActionListener**. The calculator performs addition and subtraction, while the bank balance application performs deposit and withdrawal operations. These programs show how Java Swing can be used to create **interactive and user-friendly GUI applications**.